

- Q.25 Explain types of fault and remedies, that can occur in battery ignition system.
- Q.26 What are the desirable characteristics of lubricant?
- Q.27 Define
- Indicated thermal efficiency
 - Volumetric thermal efficiency along with their formulae
- Q.28 What is the necessity of water cooling system.
- Q.29 Explain mist lubrication system.
- Q.30 Explain prony brake dynamometer with the help of neat sketch.
- Q.31 What are the methods of reducing pollution of IC. Engines?
- Q.32 Explain parallel flow jet condenser.
- Q.33 Give comparison of impulse and reaction turbine.
- Q.34 Write a short note on jet propulsion engine.
- Q.35 Write the advantages and disadvantages of ram jet engine?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain Four Stroke SI engine in detail.
- Q.37 Explain magneto ignition system with the help of neat sketch.
- Q.38 What is cooling tower? What are its different types? Explain.

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**4th Sem / Mech, CNC, Mech. Adv. Manuf. Tech.,
Mech Engg (CAD/CAM Dsgn & Robotics)**

Subject:- Thermodynamics - II. / I.C. Engines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Two strokes petrol engine is used in
- Scooters
 - Two wheelers
 - Moped
 - All of the above
- Q.2 The function of ignition coil in battery ignition system is to
- increase the current
 - increase the voltage
 - reduce sparking
 - all of the above
- Q.3 Which of the following is not a type of nozzle
- Pintaux nozzle
 - Pintle nozzle
 - Single hole nozzle
 - Surface nozzle
- Q.4 The compression ratio of a C.I. engine is kept as
- 6:1 to 11:1
 - 16:1 to 20:1
 - 1:1 to 5:1
 - All of the above

- Q.5 Pressurized cooling system is used for
- heavy duty engines
 - stationary engines
 - scooters
 - none of the above
- Q.6 Overcooling of engine may cause
- Failure of lubrication
 - decrease in friction losses
 - reduction in strength of parts
 - starting problem of engines
- Q.7 Planimeter is used to find
- Brake power
 - Indicative power
 - Friction power
 - All of the above
- Q.8 Delaval turbine is
- Impulse turbine
 - Reaction Turbine
 - Both a and b
 - Neither a nor b
- Q.9 Mechanical efficiency of gas turbine is always
- >1
 - <1
 - $=1$
 - zero
- Q.10 The ram jet engine has
- single compression unit
 - double compression unit
 - no compression unit
 - Either a or b

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Carburation .
- Q.12 The compression ratio of petrol engine varies from_____.
- Q.13 Fuel injector and fuel pump is used is_____ engine.
- Q.14 Radiators are usually made up of_____.
- Q.15 A lubricant should have _____ point.
- Q.16 The full form of LPG is_____.
- Q.17 Brake power is measured by_____.
- Q.18 When the cross sectional area of the nozzle decreases continuously from entrances to exit it is called_____.
- Q.19 Expand CRDZ.
- Q.20 Constant volume gas turbine works on_____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 The compression ratio of an I.C. engine operating on Otto cycle is 8. The specific heat ratio of the working fluid is 1.4. Calculate the thermal efficiency.
- Q.22 Write a comparison between internal combustion engine and external combustion engines.
- Q.23 Explain the types of carburetors on the basis of direction of flow.
- Q.24 Explain multi point fuel injection system.